DATA 606 CAPSTONE · FINAL PRESENTATION

Predicting Movie Box-Office Success Before Release

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BACKGROUND

A multi-million-dollar bet, made on incomplete information

- Studios commit tens to hundreds of millions of dollars years before a single ticket is sold.
- Much of what decides a film's fate — reviews, word of mouth, competition — only appears after release.
- But real signals are known beforehand: budget, genre, runtime, release timing, language, studio.
- **Goal: predict a film's financial success from pre-release attributes alone.**

THE DELIVERABLE

An end-to-end pipeline: data → EDA → machine-learning models → an interactive app that scores a film's odds of success.

THE TARGET

What counts as “success”?

$$\text{Success} = \textbf{Revenue} \geq \textbf{2} \times \textbf{Budget}$$

- Marketing roughly doubles the cost — studios spend ~50–100% of the budget again on advertising.
- Theaters keep about half the box-office gross, so studios net only ~50% of reported revenue.
- **→ A film must gross about twice its budget just to break even.**

56%

of films clear the bar — a balanced target, ideal for classification.

THE DATA

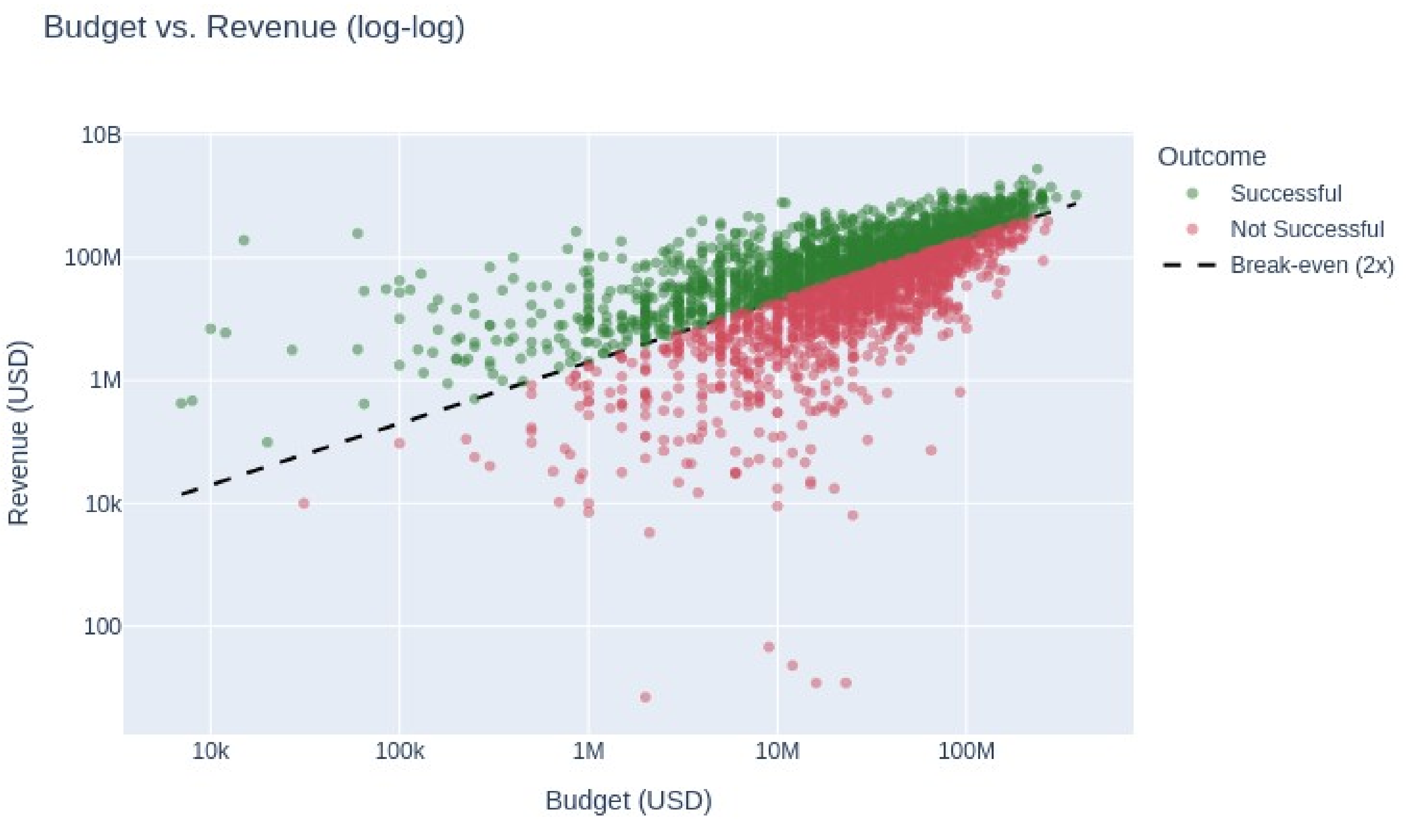
TMDB 5000 Movie Dataset

4,803	3,215	20	1916-2016
movies (raw)	usable after cleaning	original columns	release years

- Each row is one movie: budget, revenue, genres, runtime, release date, language, studios.
- Cleaning: released films only; plausible budget (≥ $1,000 — 13 films had impossible $1–$250 budgets) and real revenue; parse JSON fields.
- Engineered features: ROI, season, release window, genre-timing flags, sequel/franchise flag, major-studio flag.

Source: The Movie Database (TMDb), via Kaggle.

Budget vs. revenue — the break-even line



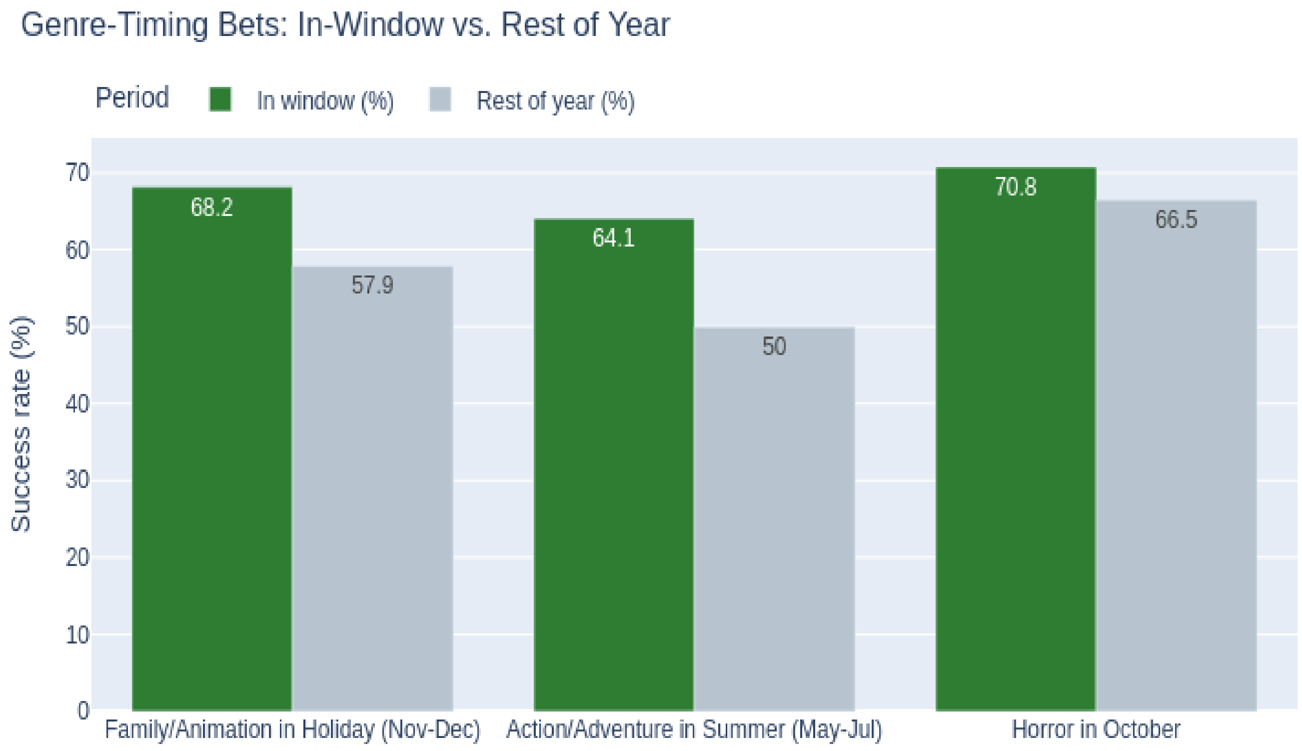
- Dashed line = break-even (revenue = 2× budget). Green cleared it; red didn't.
- Bigger budgets earn more in absolute terms — but plenty of expensive films still miss.
- **Budget matters, but it isn't destiny — which is why this is worth modeling.**

EDA HIGHLIGHTS

When a film opens matters

- Success peaks in June (~67%); September is worst (~43%) — and one of the busiest: the “dump month.”
- By industry window: Summer and Holiday ~61% vs. 56% baseline; early Fall just 47%.
- Genre matters too: horror leads (~78%), action is lowest (~50%).

The right genre in the right window



- Action/Adventure in summer: 64% vs 50% the rest of the year (+14 pts).
- Family/Animation over the holidays: 68% vs 58% (+10 pts).
- Horror clusters in October — its single biggest release month.
- **These patterns become features the model can use.**

Approach — strictly pre-release features, no leakage

16 input features, all knowable before release

- Financial: budget (log) · spend intensity (budget per minute)
- Content: primary genre · runtime · # genres · language(s)
- Timing: release year · season · industry release window · genre-timing flags
- Production: # production companies · major-studio flag · sequel/franchise flag

Deliberately excluded (leakage)

- Revenue, ROI, profit — the outcome itself.
- Popularity, ratings, vote counts — these only exist after release.
- **The model only sees what a studio would know at the greenlight moment.**

Setup: stratified 80/20 train-test split (2,572 / 643 films) + 5-fold cross-validation · three classifiers: Logistic Regression, Random Forest (tuned), Gradient Boosting.

MACHINE LEARNING

Feature engineering — two signals that moved the needle

SEQUEL / FRANCHISE

68% vs 54%

Detected from plot keywords (sequel, superhero, based on novel…) and title patterns (Part, 2, III…). Franchise films clear the bar far more often.

MAJOR STUDIO

61% vs 51%

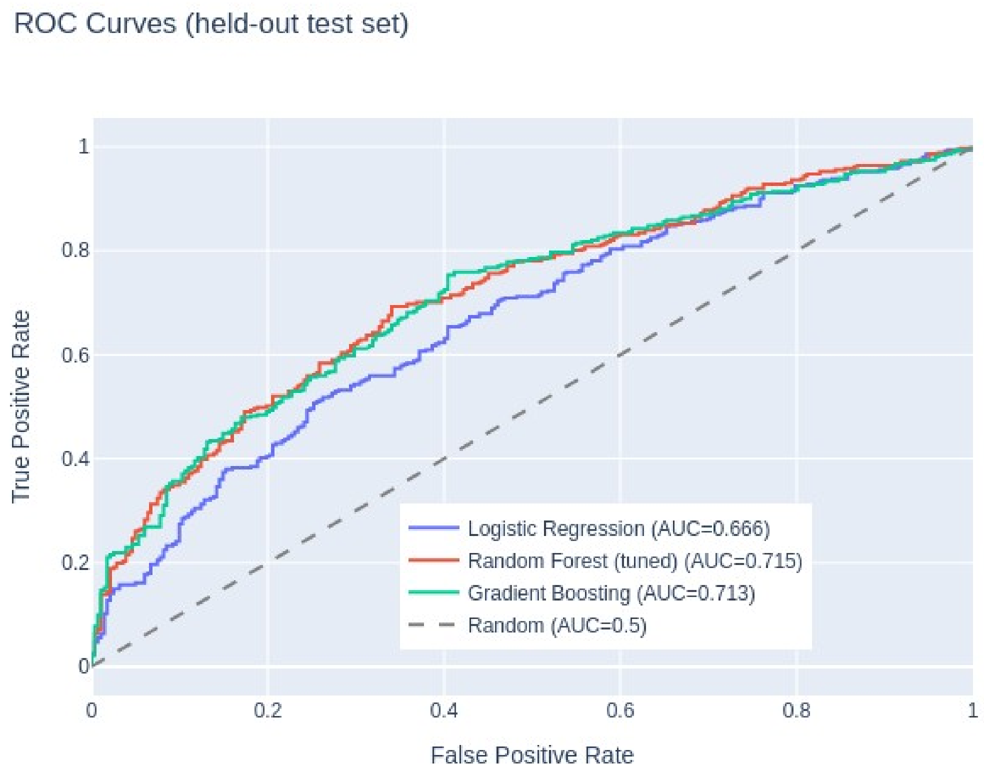
Whether a major (Disney, Warner Bros., Universal…) produced the film — distribution muscle and marketing reach show up in outcomes.

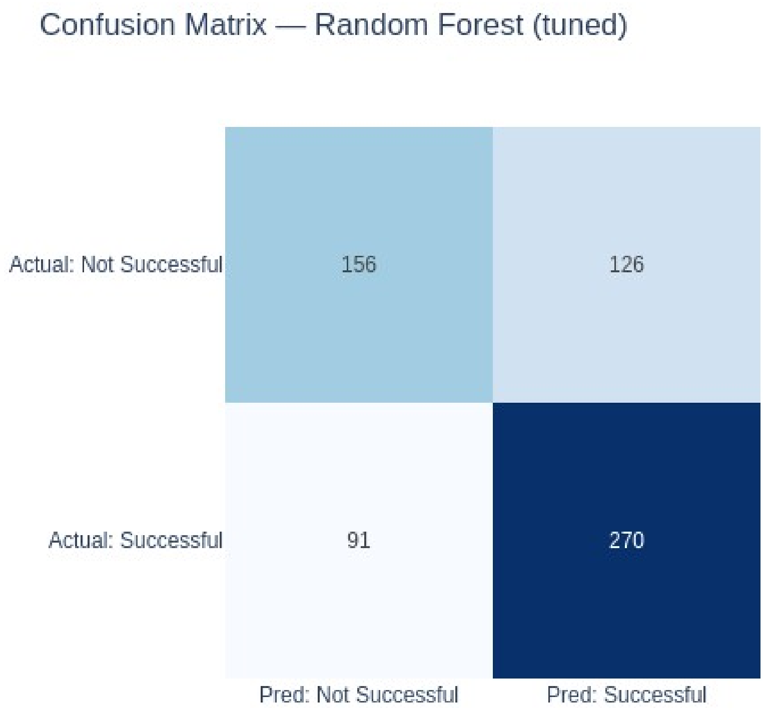
Also added: release year (era effects) and spend intensity (budget ÷ runtime). Together these lifted test ROC-AUC from 0.66 to 0.71.

Results — all models beat the baseline

- **Best held-out test: 68% accuracy, 0.715 ROC-AUC (baseline: 56%).**
- Gradient Boosting and the tuned Random Forest are essentially tied; Logistic Regression trails.
- High recall (~0.75): the model catches most true successes.

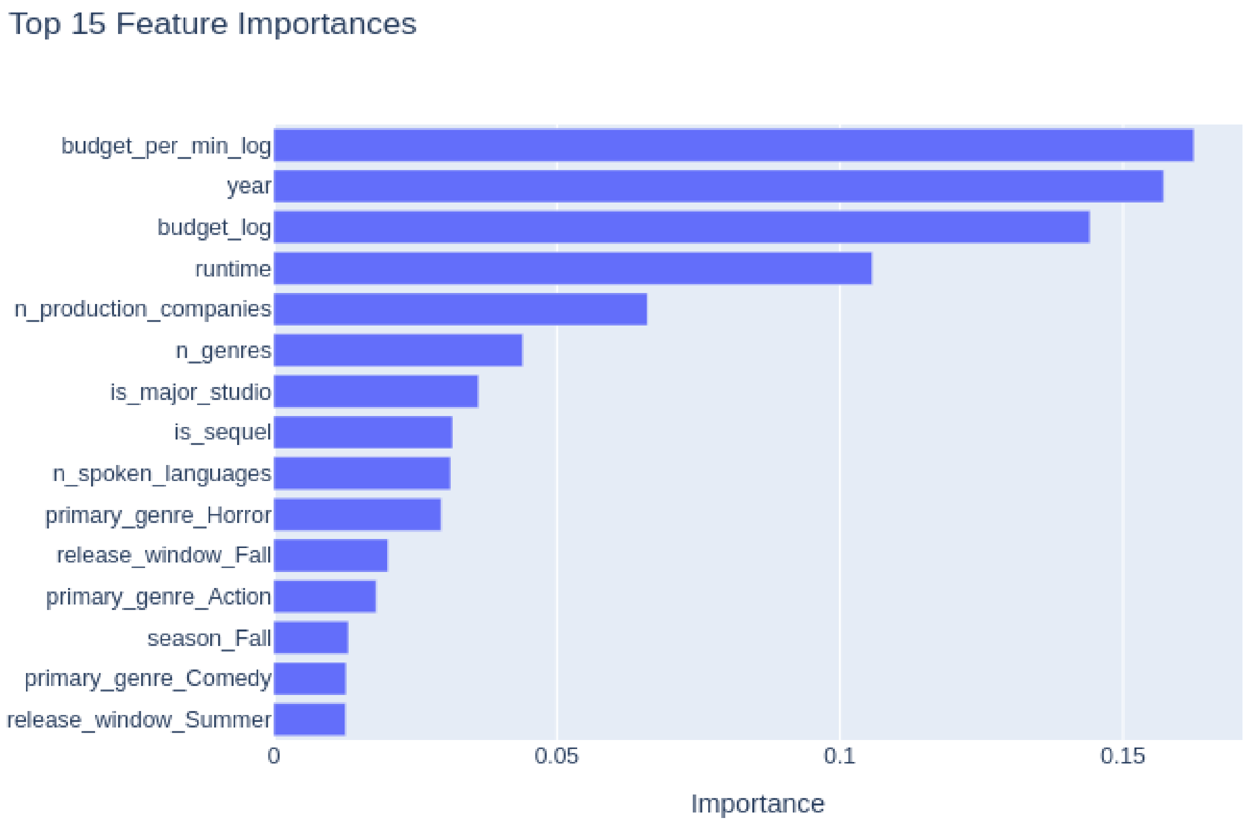
Robustness — cross-validation and error profile





5-fold cross-validation: tuned Random Forest is most robust — 64% accuracy, 0.683 AUC (±0.025). A single split can flatter a model; CV keeps us honest.

What drives the predictions



- Financial scale dominates: spend intensity, release year, and budget are the top three.
- Runtime, studio count, and the sequel / major-studio flags follow.
- Genre (Horror) and the timing features add real secondary signal — the EDA and the model agree.

Streamlit app — every input, explained

What the user enters

- Budget — the strongest single predictor (log-scaled internally).
- Primary genre — horror/animation outperform; action is hardest.
- Runtime — also feeds spend intensity (budget ÷ minute).
- Release month & year — timing and era effects.
- Language · # production companies (+ advanced: # genres, # languages).
- Sequel/franchise ✓ and major-studio ✓ — the two engineered flags.

What the app does

- Derives the hidden features automatically — season, release window, genre-timing flags, spend intensity.
- Trains the most robust model (tuned Random Forest) on the full cleaned dataset.
- Returns a probability of success with a clear verdict — plus the honest caveat that the 2× target is a proxy for profit.
- **One command to run: streamlit run app/app.py**

LIVE DEMO

Movie Box-Office Success Predictor — running locally at localhost:8501

If the demo gods are unkind: *summer franchise tentpole → 72% · low-budget October horror → 72% · big-budget September drama → 48%.*

CONCLUSION

Takeaways, limitations, and what's next

Takeaways

- Pre-release attributes predict success meaningfully better than chance.
- Budget dominates; genre, timing, and franchise/studio signals add real lift.

Limitations

- 2× rule is a proxy — revenue is gross; budget excludes marketing.
- Data ends in 2016 and skews to larger films; ~33% of rows lacked usable financials.

Future work

- Star power, marketing spend, and competition features.
- Newer data + streaming; NLP on plot summaries; ROI-tier prediction.

Predicting movie success before the cameras roll.

Bennett Speicher · github.com/bdspeicher12/UMBC-DATA606-Capstone

Thank you — questions welcome.